TEXORA · COURSE NOTES · LECTURE SHEET

Cellulosic Fibres: Cotton, Flax, Jute, Ramie and Hemp

The plant fibres built from cellulose — where each one grows, how it is extracted, its properties, and the work each fibre does in the world.

TE-1209 · NATURAL TEXTILE FIBRES · TOPIC 2 · LEVEL 1 TERM 2

1. The common chemistry: cellulose

All five fibres of this topic are built mainly from **cellulose**, the structural polymer of plant cell walls — a long chain of glucose units joined end to end, thousands of units long in a mature cotton fibre. Because the chemistry is shared, the family behaviour is shared, and it is worth learning once:

- **Water:** cellulose carries many hydroxyl (–OH) groups that attract water, so all cellulosic fibres absorb moisture readily, swell when wet, and feel comfortable against skin.
- **Acids damage them:** strong acids hydrolyse the chain and tender (weaken) the fibre; even dilute acid left to dry on cotton can rot it.
- **Alkalis are tolerated:** cellulosics resist alkali well — cotton is deliberately treated with strong caustic soda in mercerisation to gain lustre and strength, something no protein fibre would survive.
- **Heat and fire:** they do not melt; they scorch, char and burn like paper, and are degraded by prolonged strong sunlight.
- **Biology:** mildew and bacteria feed on cellulose in warm, damp storage — a permanent storage concern.

Within the family, the fibres divide by the part of the plant they come from: **seed fibres** (cotton), which grow as hairs on a seed, and **bast fibres** (flax, jute, ramie, hemp), which are the bundles in the stem that hold the plant upright. That single difference in origin drives almost everything else in this topic — extraction method, fibre length, fineness and use.

2. Cotton — the seed fibre that runs the industry

Cotton is the fibre of the seed hairs of the cotton plant (*Gossypium* species). When the boll ripens it bursts open to reveal a mass of white fibres, each a single elongated cell grown from the seed coat. The fibres are picked (by hand or machine), the seeds are removed by **ginning**, and the lint is pressed into

bales for shipment to spinning mills. Cotton alone accounts for roughly a quarter of world fibre use and remains the reference fibre against which others are judged.

2.1 Structure and properties

Under the microscope a cotton fibre is a flat, twisted ribbon — the twists, called **convolutions**, appear as the fibre's thin wall collapses on drying. The cross-section is kidney- or bean-shaped with a central hollow (the *lumen*). Properties that matter in practice:

- **Length:** roughly 12–50 mm by variety — long-staple types (Egyptian, Pima) at the top, short coarse types at the bottom. Length is the first determinant of the fineness of yarn that can be spun.
- **Strength:** moderate, and unusual in being *stronger wet than dry* (by roughly a tenth to a fifth) — one reason cotton survives a lifetime of washing.
- **Absorbency:** moisture regain about 7–8.5%; absorbs water readily, dries slowly, dyes well with the cotton dye classes.
- **Elasticity:** low — cotton creases easily and stretches little, recovering poorly; this is why cotton garments are blended or finished for easy care.
- **Handle:** soft, cool, non-irritating; no static problems because of its moisture content.

Uses: the widest of any fibre — apparel of every kind, denim, shirting, towels and bed linen (absorbency), sewing thread, medical cotton, and blends with polyester and elastane.

3. The bast fibres: how a stem becomes a fibre

In a bast plant, the fibres lie in bundles in the stem's bark region, glued to the woody core and to each other by pectins and (in jute and hemp) lignin. Freeing them is a biological, not a mechanical, first step — **retting**: the harvested stems are spread on fields (dew retting) or submerged in ponds and slow water (water retting), and bacteria digest the pectin so the fibre bundles can be separated from the straw. The chain for flax shows all the steps:



Figure 1 — From flax plant to yarn: retting frees the fibre bundles, breaking and scutching remove the woody core, and hackling combs out the long "line" fibres from the short "tow".

After retting, the straw is **broken** (crushed between rollers), **scutched** (beaten to shake out the woody fragments, called shives), and **hackled** (drawn through beds of pins that comb the bundles parallel and separate the long, valuable *line* fibres from the short *tow*). Jute follows the same biological logic with

simpler mechanics — retted stems are stripped by hand and the strands washed and dried — because its end uses tolerate a coarser fibre.

4. Flax (linen)

Flax fibres are long (the ultimate cells commonly 20–40 mm, and the technical fibre — the bundle as spun — much longer), smooth, and lustrous, with characteristic cross-markings (nodes) along their length. Flax is stronger than cotton, especially wet, highly absorbent yet quick to dry, and cool to the touch because it conducts heat away from the body — the reason **linen** (the name of flax yarn and fabric) is the classic summer clothing fibre. Its weaknesses are low elasticity and resilience: linen creases sharply and recovers badly. Uses: summer apparel, handkerchiefs, table linen, canvas, sewing thread, and increasingly composite reinforcement.

5. Jute — the golden fibre

Jute, the bast fibre of *Corchorus* plants, is Bangladesh's heritage fibre and the country is the world's leading exporter of raw jute. Chemically jute is only partly cellulose: it carries a significant proportion of **lignin** (roughly an eighth of its weight), which stiffens the fibre, colours it golden-brown, and makes it harsher and more brittle than flax. The individual cells are short (a few millimetres), so jute is always used as a long technical strand of cemented cells. Properties follow: strong in the strand, cheap, coarse, highly absorbent at first but weakened by prolonged damp, and biodegradable in months. Classic uses are **hessian (burlap), sacking and carpet backing** — the world's packaging fibre — and a growing range of diversified products: geotextiles for erosion and road work, shopping bags, decorative fabrics, and fibre-reinforced composites.

> **WORKED REASONING — WHY JUTE CANNOT SIMPLY REPLACE COTTON IN APPAREL**
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> A fine shirt yarn needs fibres that are long relative to their fineness, flexible, and uniform. Jute fails on structure, not on strength: its ultimate cells are only a few millimetres long and are cemented into stiff bundles by lignin, so jute cannot be drafted into fine, soft, even yarn on cotton machinery, and its harshness and low flexibility would make an unwearable fabric. Jute's excellence lies where coarseness is acceptable and biodegradability is a virtue — sacks that rot with their contents' journey done, geotextiles that hold a slope until grass takes over. Matching the fibre to the duty, not ranking fibres absolutely, is the professional habit this example teaches.

6. Ramie and hemp

Ramie (China grass) is the aristocrat of the bast fibres: very long, among the strongest of all natural fibres, brilliantly white and lustrous, and highly absorbent. Its gum content is not removed by retting and must be dissolved chemically (*degumming*), and the fibre is stiff with low elasticity, so ramie is often blended; uses include fine fabrics, canvas, and paper-making.

Hemp is coarse, exceptionally strong and durable, resistant to rot and mildew, and long used for ropes, sailcloth and canvas — the word *canvas* itself derives from *cannabis*. Modern hemp is bred to negligible narcotic content and is returning in apparel blends, footwear and composites on its durability and low-input cultivation.

7. The five compared

FIBRE	PLANT PART	DISTINGUISHING CHARACTER	TYPICAL USES
Cotton	Seed hair	Soft, absorbent, stronger wet; the general-purpose fibre	Apparel, denim, towels, bed linen, thread
Flax	Stem (bast)	Long, lustrous, cool; creases badly	Linen apparel, table linen, canvas
Jute	Stem (bast)	Coarse, golden, lignified, biodegradable	Hessian, sacking, geotextiles, bags
Ramie	Stem (bast)	Longest and strongest of the group; stiff, needs degumming	Fine fabrics (blended), canvas, paper
Hemp	Stem (bast)	Coarse, very durable, rot-resistant	Rope, canvas, workwear blends, composites

8. How cotton quality is measured and traded

Because cotton is a natural product, no two bales are identical, and the spinning mill buys properties, not just weight. The measurements that appear on every cotton contract and classing report are:

- **Staple length** — the characteristic length of the longer fibres in the sample, measured by instruments (the digital fibrograph principle: scanning a combed beard of fibres). Length classes run from short (under about 20 mm) through medium and long to extra-long staple. Longer cotton spins finer, stronger yarn with less waste.

- **Fineness and maturity — the micronaire value.** A fixed weight of cotton is compressed in a chamber and air is blown through it; the airflow depends on the fibres' surface area, giving a single number that combines fineness and maturity. Very fine-but-immature cotton is a dyeing hazard (it forms pale specks called neps and dye-resist spots), while coarse cotton limits yarn fineness.
- **Strength** — the breaking force of a bundle of fibres (measured by Stelometer-type instruments), reported so mills can predict yarn strength.
- **Trash and grade** — the proportion of leaf, seed-coat fragments and dust, together with colour grade (white, spotted, tinged), judged against official standards; trash becomes waste and cleaning loss in the blowroom.
- **Uniformity** — the spread of fibre lengths; a sample with many short fibres spins badly however good its average length, so short-fibre content is reported separately.

Modern classing (the HVI — High Volume Instrument — line) measures all of these in under a minute per sample, and a mill's buying and mixing (laydown) decisions are made from the printout. The textile engineer's first quantitative skill is reading that report and foreseeing what the cotton will do in the mill.

8.1 A processing note: mercerisation

One treatment belongs with this topic because only cellulose fibres can receive it. When cotton yarn or fabric is held under tension in strong caustic soda (about 18–25% NaOH) and then washed, the fibres swell, their cross-section rounds out, and the material gains permanent **lustre, strength (roughly a fifth more), and dye affinity** — the process of **mercerisation**, met again in wet-processing courses. It works because alkali penetrates and rearranges the cellulose structure without dissolving it — the family property of §1 turned into a commercial finish.

9. Glossary — key terms of this topic

TERM	MEANING
Ginning	Separating cotton lint from the seed after picking.
Retting	Rotting the pectin that binds bast fibres to the stem, by dew or water and bacterial action.
Scutching	Beating retted straw to remove the broken woody core from bast fibres.
Hackling	Combing bast fibres through pins to parallel them and separate line from tow.
Ultimate cell	A single plant cell; the textile "fibre" of bast plants is a strand of many ultimate cells.

Lignin	The stiffening substance in plant stems; its presence makes jute and hemp coarser than pure cellulose fibres.
Degumming	Chemical removal of the natural gums from ramie (and of sericin from silk).

10. Key points and summary

- Cellulosic fibres share cellulose chemistry: absorbent, alkali-resistant, acid-sensitive, and burning like paper. Cotton is a seed fibre; flax, jute, ramie and hemp are bast (stem) fibres.
- Cotton — soft, absorbent, stronger when wet, low in elasticity — is the general-purpose fibre; its length grades decide the yarns it can spin.
- Bast fibres are freed by retting, then broken, scutched and hackled; the lignin in jute and hemp trades fineness for cheapness, stiffness and biodegradability.
- Each fibre is best somewhere: linen for cool summer cloth, jute for packaging and geotextiles, ramie and hemp for strength and durability.

11. Practice

Answer in your own words; brief model answers follow.

1. State the chemical reason why all five fibres of this topic behave alike toward acids and alkalis.
2. Cotton is described as “stronger wet than dry”. Why is this property commercially valuable, and which common product depends on it daily?
3. Arrange cotton, flax, jute and ramie in order of fibre fineness (finest first), and justify the position of jute using its structure.
4. A mill receives two cotton lots: Lot A — staple 32 mm, micronaire 4.2; Lot B — staple 24 mm, micronaire 5.4. Which lot would you choose for a fine shirting yarn, and why?
5. Explain each step of the flax extraction chain (Figure 1) in one sentence, naming what is removed at that step.
6. Why must ramie be degummed while flax is not? What property of ramie makes the extra cost worth paying?
7. A buyer wants a fully biodegradable fabric to hold a riverbank for one season and then disappear. Choose a fibre from this topic and defend the choice against one alternative.
8. What does a high short-fibre content do to spinning, even if the average staple length is good?

Answer key

1. All are built from cellulose; acids hydrolyse the cellulose chain (tendering), while alkali is tolerated and even used (mercerisation).
2. Washing repeatedly stresses wet fabric; cotton towels, bed linen and workwear survive years of laundering partly because wet cotton gains strength instead of losing it (unlike wool or viscose).
3. Ramie/flax finest as technical fibres at their best, then cotton, jute last — jute's "fibre" is a lignin-cemented bundle of short cells, inherently coarse.
4. Lot A: longer staple spins finer and stronger yarn; micronaire 4.2 indicates mature, medium-fine fibres. Lot B is short and coarse — suited to coarse yarns only.
5. Harvest (plant removed), retting (pectin digested), breaking & scutching (woody shives removed), hackling (short tow separated from line), spinning (bundles drafted into yarn).
6. Ramie's gums are not pectin removed by retting; they need chemical dissolution. Its exceptional length, strength and lustre repay the cost.
7. Jute (or coir) geotextile: strong enough for a season, cheap, and biodegrades in months; cotton would be far too costly and less rot-suited to wet soil work — jute is the standard choice.
8. Short fibres are uncontrolled in drafting: they raise ends-down, yarn unevenness and hairiness, and are lost as waste — uniformity matters as much as mean length.
